

Project Design Phase-II Technology Stack (Architecture & Stack)

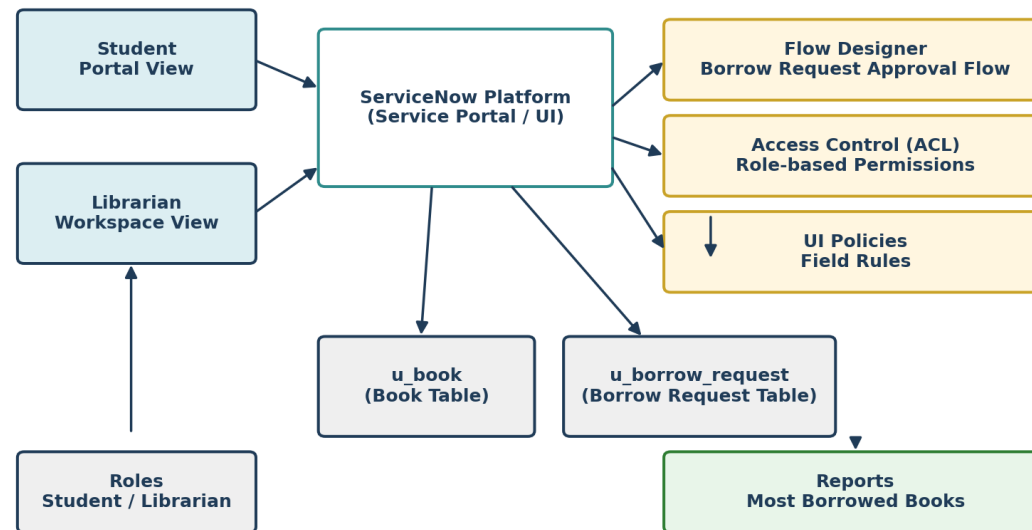
Date	05 July 2026
Team ID	Solo Project – Individual Capstone
Project Name	Library Management System (ServiceNow)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Figure 1: Library Management System - Solution Architecture (ServiceNow)



Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How students & librarians interact with the application	ServiceNow Service Portal / native platform UI forms
2.	Application Logic-1	Approval and status-update logic for borrow requests	Flow Designer (Borrow Request Approval Flow)
3.	Application Logic-2	Field-level usability rules (mandatory Return Date on Issue)	UI Policies
4.	Application Logic-3	Restricts book selection to available copies only	Reference Qualifier (Simple, on Book reference field)
5.	Database	Stores catalog and transaction data	ServiceNow native tables (u_book, u_borrow_request)
6.	Cloud Database	Platform-managed backend data store	ServiceNow-hosted database (Personal Developer Instance)
7.	File Storage	Not required for current release (no attachments)	Not applicable
8.	External API-1	Not used in current scope	Not applicable
9.	External API-2	Not applicable in current scope	Not applicable
10.	Machine Learning Model	Not applicable in current scope	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployed and tested on a ServiceNow Personal Developer Instance (PDI). Local Server Configuration: Not applicable (cloud-hosted PDI) Cloud Server Configuration: ServiceNow-managed multi-instance cloud	ServiceNow PDI (Cloud)

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Open-Source Frameworks	None — built entirely on ServiceNow's proprietary low-code Now Platform	Now Platform (Flow Designer, Service Portal, GlideRecord)
2.	Security Implementations	Role-based Access Control Lists (ACLs) restrict Create/Read/Write/Delete per role; Reference Qualifier prevents invalid data entry	ServiceNow ACL engine, role-based security
3.	Scalable Architecture	Modular custom-table design (u_book, u_borrow_request) allows new related services to be added without re-architecture	ServiceNow table extension model
4.	Availability	Hosted on ServiceNow's cloud infrastructure with platform-managed uptime/SLA	ServiceNow multi-instance cloud
5.	Performance	Reference Qualifier limits records returned to forms; reports pre-aggregate data via Group By/Count	ServiceNow Reporting engine, optimized GlideRecord queries